Lecture 7 – XAI 3: Saliency Maps

Responsible AI, 29.10.2026

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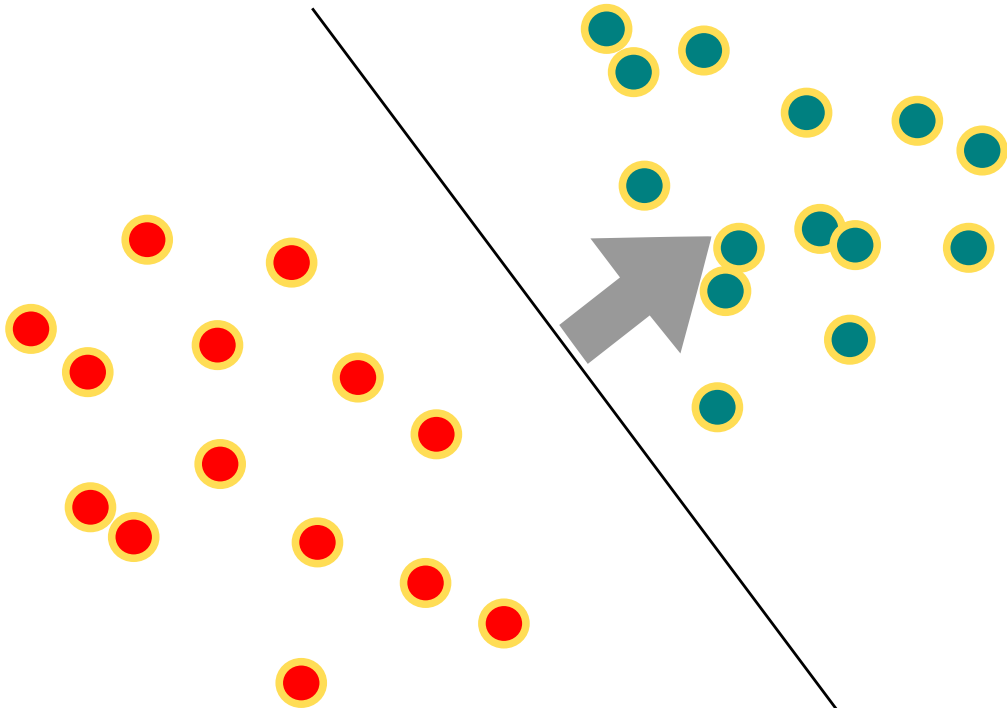
Learning goals

By the end of this lecture, you should be comfortable defining, discussing and responsibly utilizing

- Saliency maps and other feature relevance methods,
- You should have a good overview of the strengths and weaknesses of, in particular, saliency maps, and the role they play in the XAI debate.

What caused my prediction?



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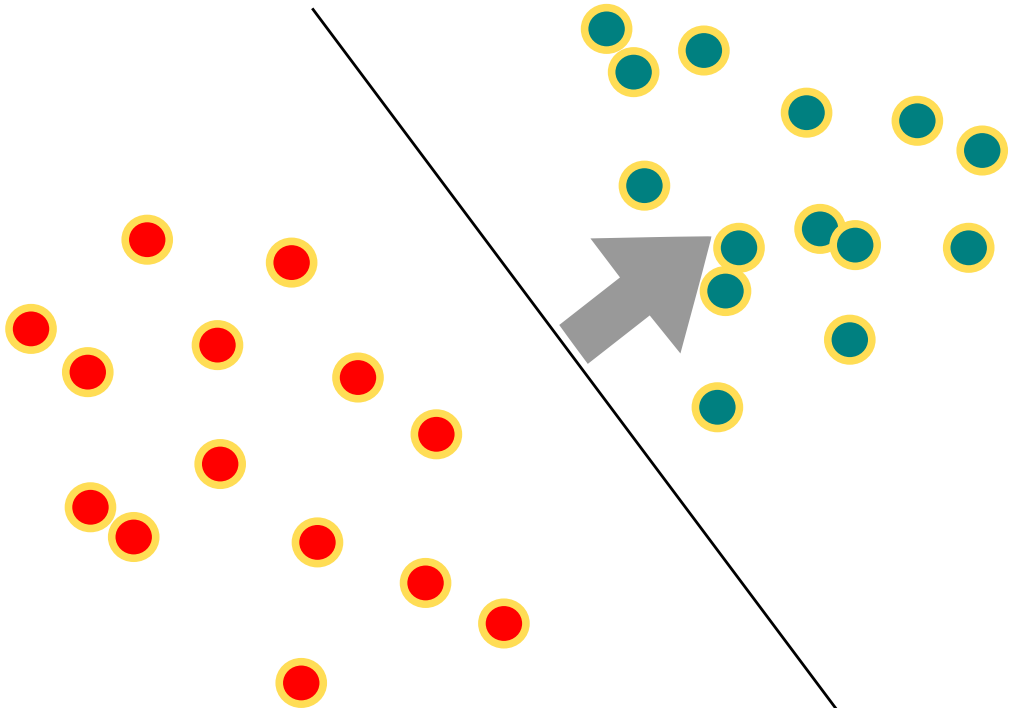
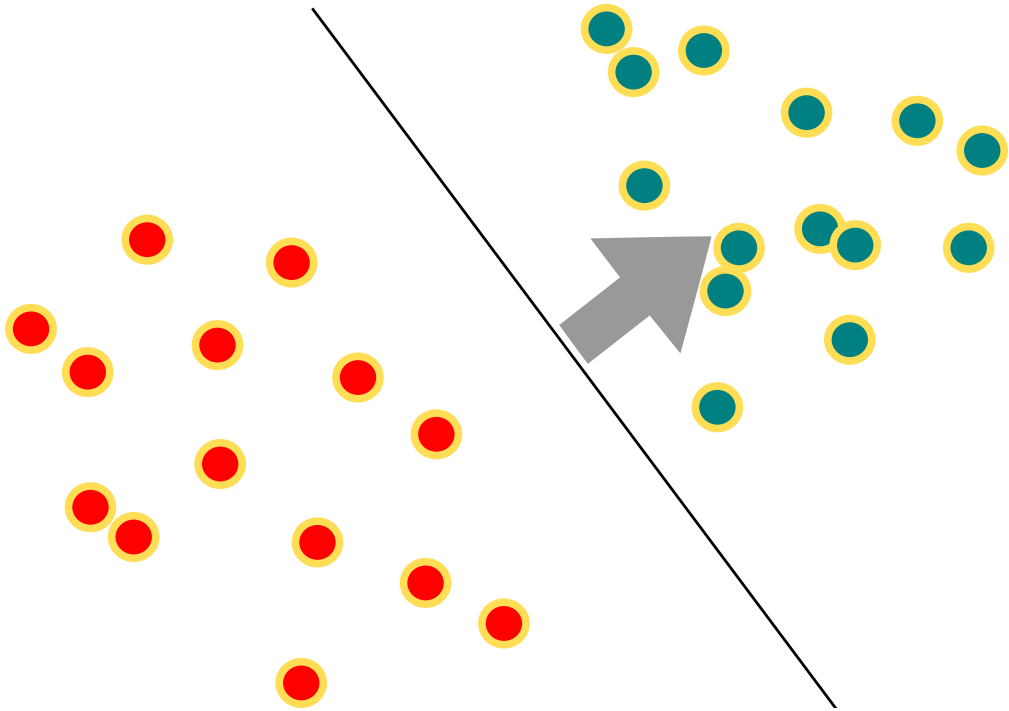


- For linear classifiers: Use the weights

What caused my prediction?



- For linear classifiers: Use the weights
- For nonlinear classifiers: Generalize via gradients

Saliency maps

Gradients

- For a function $f : \mathbb{R}^d \to \mathbb{R}$, the gradient

$$\nabla_x f = (\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \ldots, \frac{\partial f}{\partial x_d})$$

  at $x_0 \in \mathbb{R}^d$ is a vector in $\mathbb{R}^d$ pointing in the direction of maximal slope.
- For an image classifier $f : \mathbb{R}^{H \times W} \to \mathbb{R}$ predicting probability of positive class, and an image $x \in \mathbb{R}^{H \times W}$, what is ∇f?

Gradients

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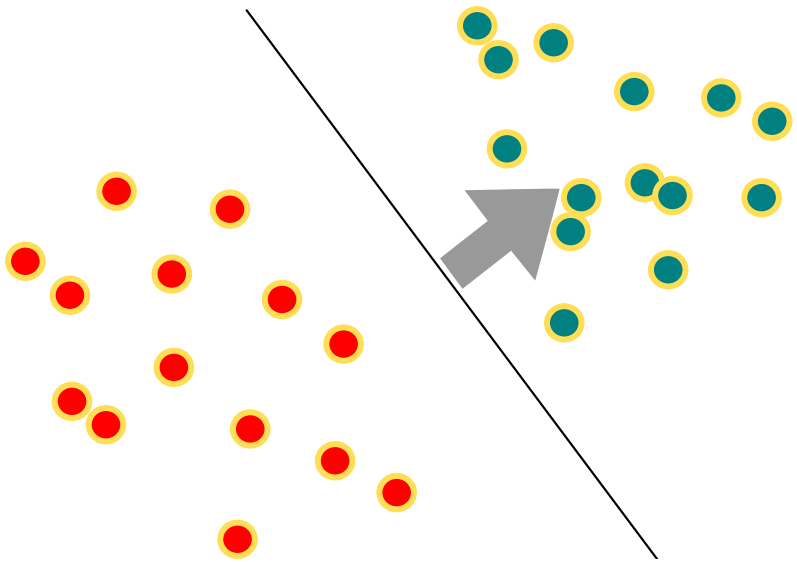
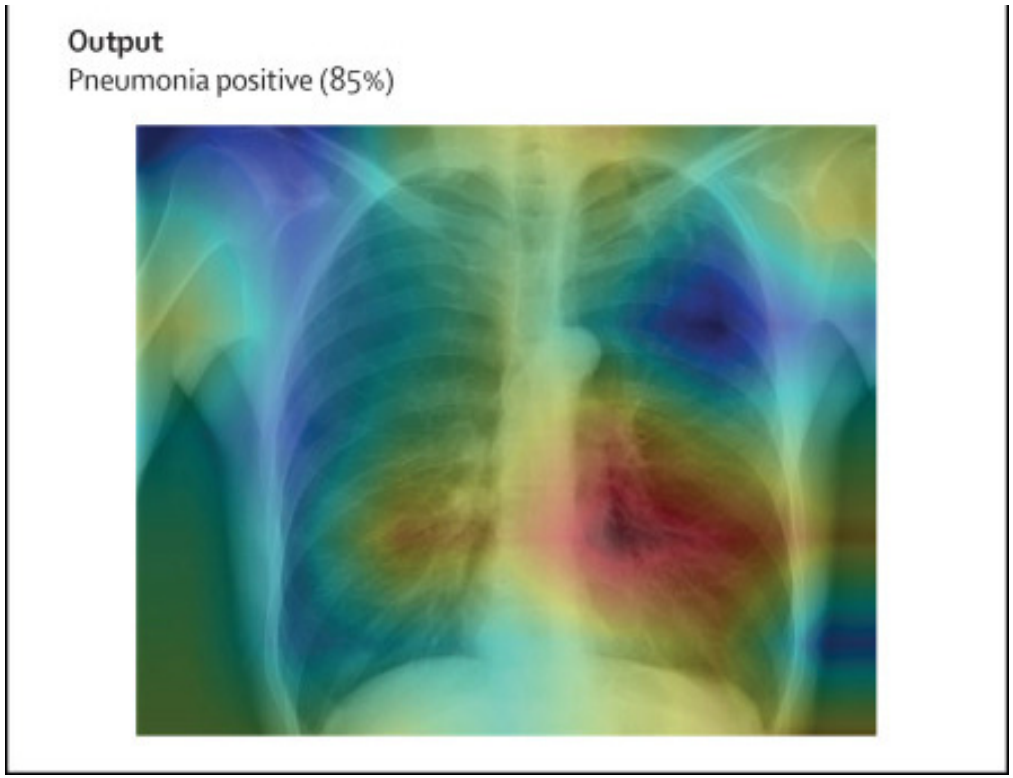
- For an image classifier $f : \mathbb{R}^{H \times W} \to \mathbb{R}$ predicting probability of positive class, and an image $x \in \mathbb{R}^{H \times W}$, what is ∇f?
- A matrix (image) in $\mathbb{R}^{H \times W}$, pointing in the direction of the positive class.



Saliency maps

A *saliency map* is a heatmap in the image plane that aims to emphasize those regions of the image that were the most important – or *salient* – for the made prediction

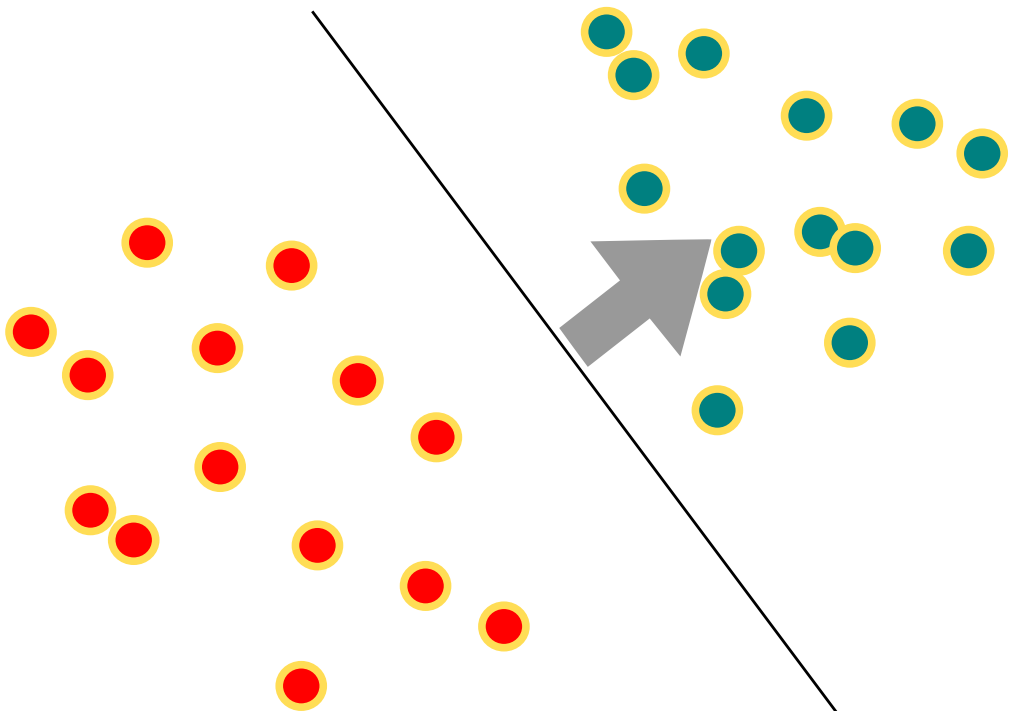


Vanilla saliency maps: Gradients

The vanilla gradient saliency map for the image is simply given by the gradient

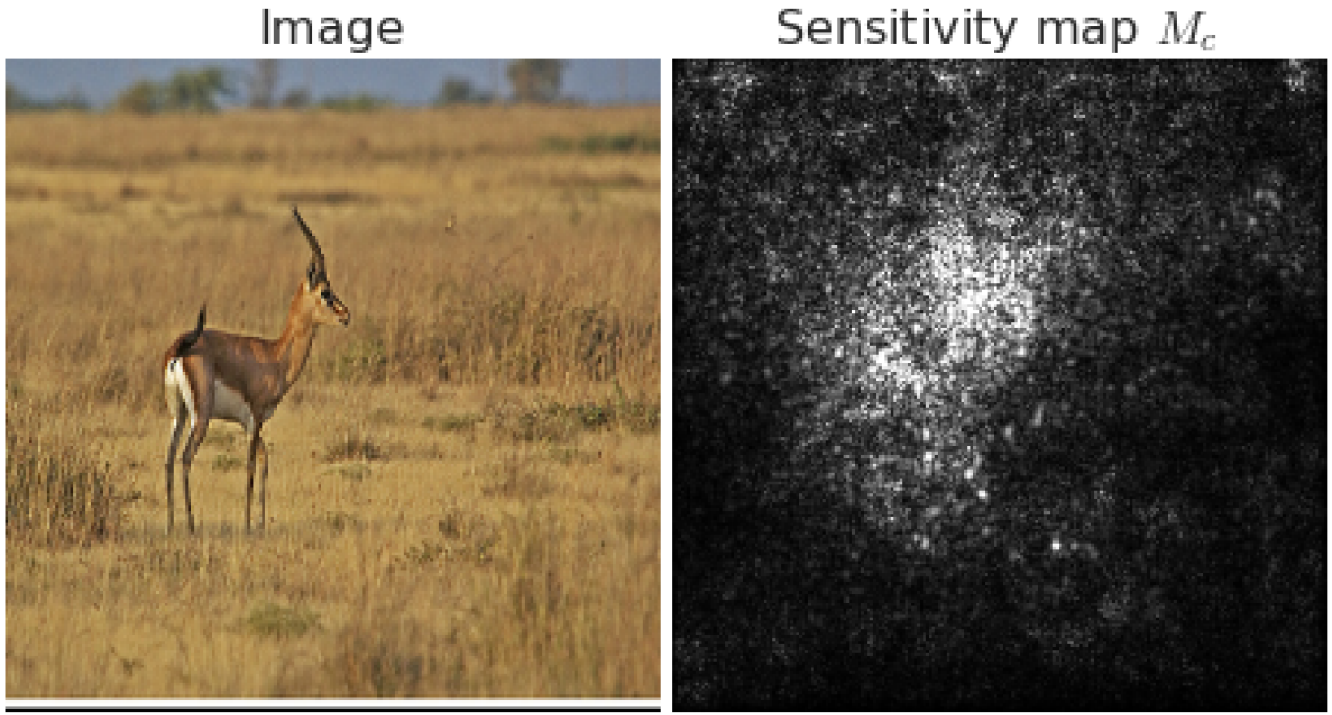
$$M_c = \nabla_x f_l(\theta, x)$$

of the (usually the correct) l^{th} class prediction of the network f at x.



SmoothGrad

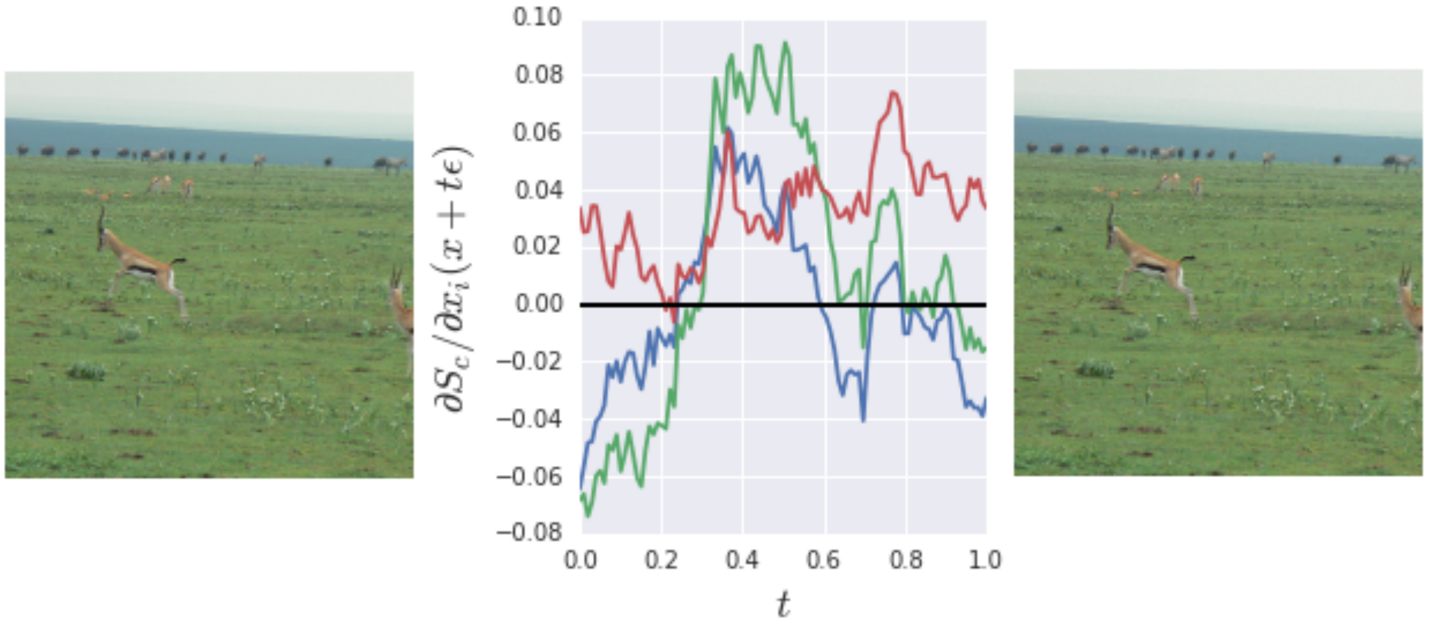
Vanilla gradients tend to look "noisy":



Smilkov et al, SmoothGrad: removing noise by adding noise, `https://arxiv.org/pdf/1706.03825.pdf`, 2017

SmoothGrad

Reason: gradients fluctuating heavily as functions on image space



Smilkov et al, SmoothGrad: removing noise by adding noise, `https://arxiv.org/pdf/1706.03825.pdf`, 2017

SmoothGrad

Smoothgrad: Apply Gaussian smoothing in image space – via stochastic approximation:

$$\hat{M}_c(x) = \frac{1}{n} \sum_n M_c(x + \mathcal{N}(0, \sigma^2))$$

where σ is the standard deviation of the Gaussian noise and M_c is the vanilla saliency map of the image x

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Smilkov et al, SmoothGrad: removing noise by adding noise, `https://arxiv.org/pdf/1706.03825.pdf`, 2017

SmoothGrad

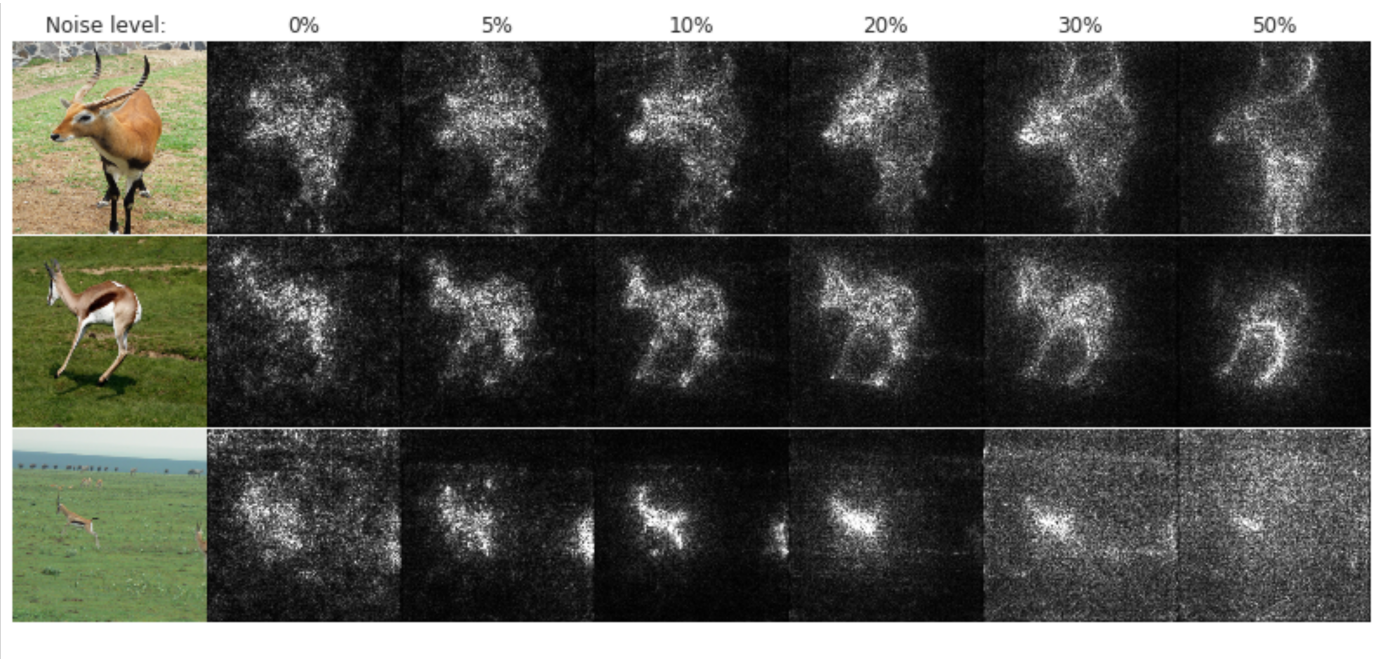

Figure 3. Effect of noise level (columns) on our method for 5 images of the gazelle class in ImageNet (rows). Each sensitivity map is obtained by applying Gaussian noise $\mathcal{N}(0,\, \sigma^2)$ to the input pixels for 50 samples, and averaging them. The noise level corresponds to $\sigma/(x_{max} - x_{min})$.

Smilkov et al, SmoothGrad: removing noise by adding noise, `https://arxiv.org/pdf/1706.03825.pdf`, 2017

Integrated Gradients

- Introduced various axioms for saliency and show that Integrated Gradients is the only method that satisfies them
- Model:

$$\text{IntegratedGrads}_i(x) ::= (x_i - x_i') \times \int_{\alpha=0}^{1} \frac{\partial F(x' + \alpha \times (x - x'))}{\partial x_i} \, d\alpha$$

- Difference from SmoothGrad: Samples images along curves rather than from a distribution

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Sundarajan et al, Axiomatic Attribution for Deep Networks, `https://arxiv.org/pdf/1703.01365.pdf`

Integrated Gradients

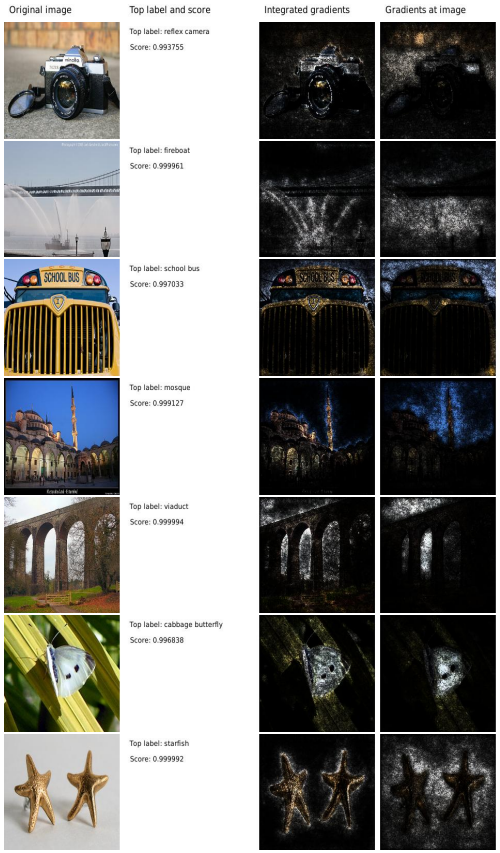


Figure 2. **Comparing integrated gradients with gradients at the image.** Left-to-right: original input image, label and softmax score for the highest scoring class, visualization of integrated gradients, visualization of gradients*image. Notice that the visualizations obtained from integrated gradients are better at reflecting distinctive features of the image.

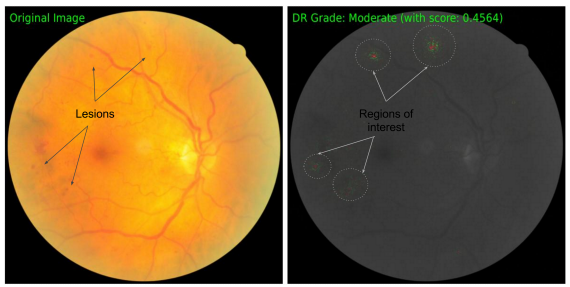
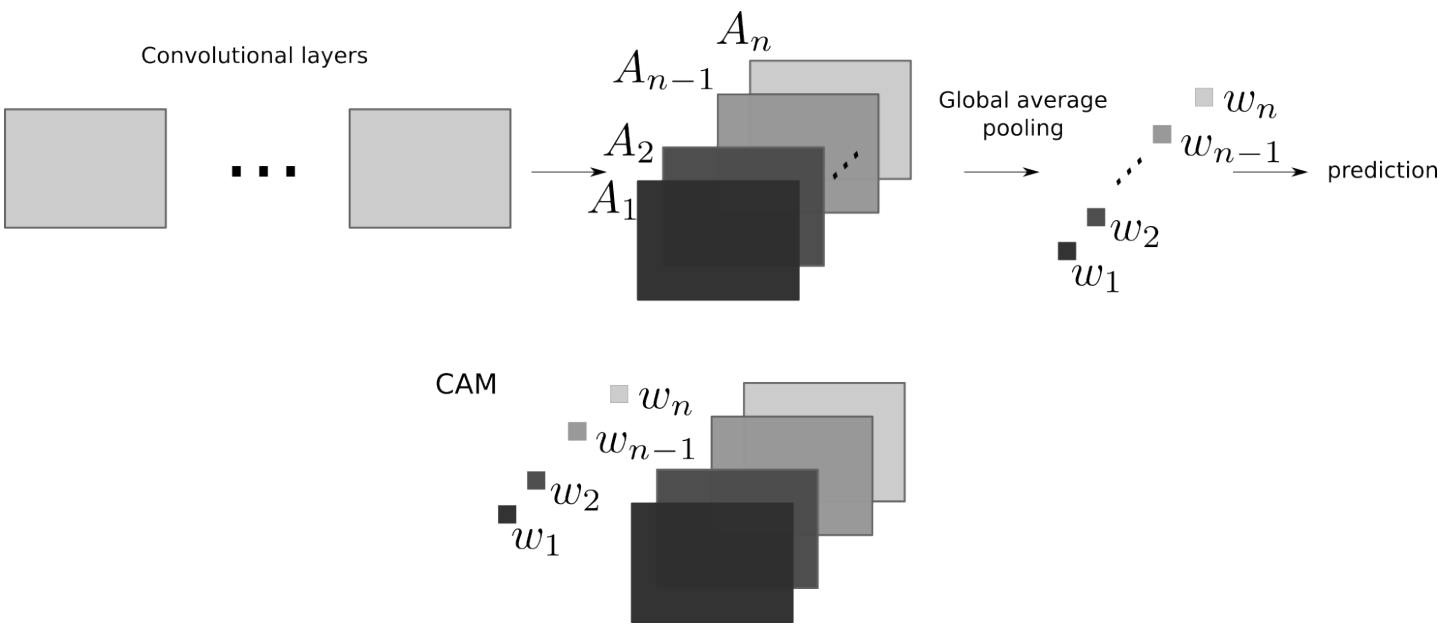


Figure 3. **Attribution for Diabetic Retinopathy grade prediction from a retinal fundus image.** The original image is show on the left, and the attributions (overlayed on the original image in gray scaee) is shown on the right. On the original image we annotate lesions visible to a human, and confirm that the attributions indeed point to them.

Sundarajan et al, Axiomatic Attribution for Deep Networks, `https://arxiv.org/pdf/1703.01365.pdf`

CAM: Class Activation Mapping



GradCAM

- Post-hoc method
- Activation mapping weights computed via gradients:

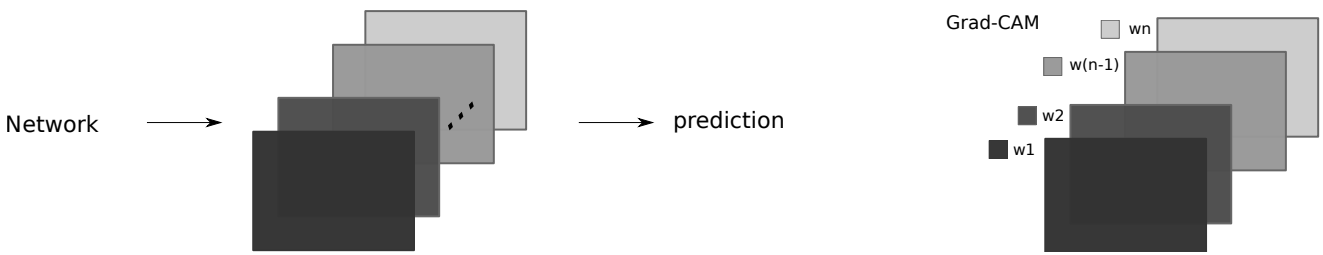
$$w_k^c = \text{mean}(\nabla_{A^k} f_c(x))$$

where $f_l(x)$ is the (before softmax) prediction of the label c for input image x, A^k is the features of the k^{th} penultimate channel.

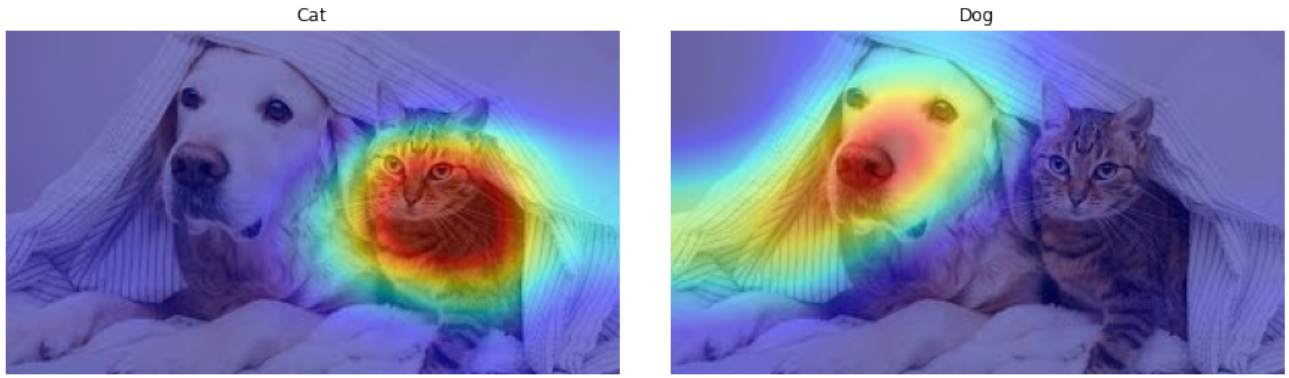
- Saliency map:

$$M_c(x) = ReLU(\sum_k w_k^c A^k)$$

- ReLU turns off negative contributions – more intuitive output



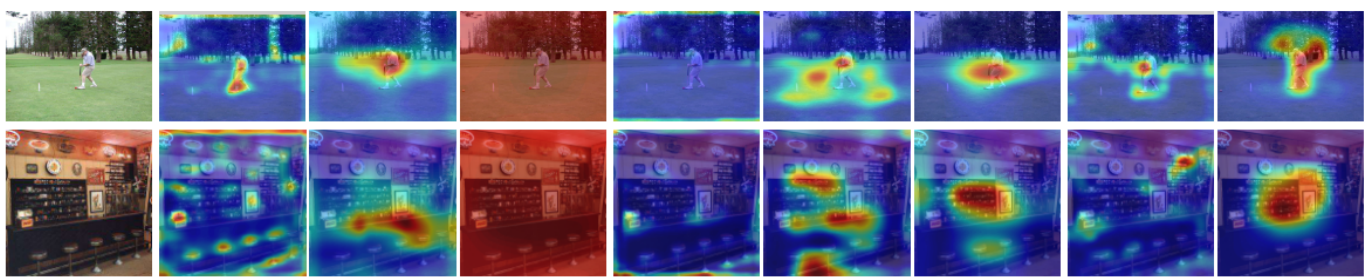
GradCAM



Examples, using cat labels vs dog labels

Saliency via attention

- The attention layer known from attention modules and transformers can also be visualized as a saliency map
- The previously encountered saliency methods were **post-hoc**: The explanation is tagged onto the method rather than embedded as an intrinsic part. Attention is **intrinsic**.



Jetley et al, Learn to Pay Attention, `https://arxiv.org/pdf/1804.02391.pdf`

Other feature relevance methods

LIME - Local Interpretable Model-Agnostic Explanations

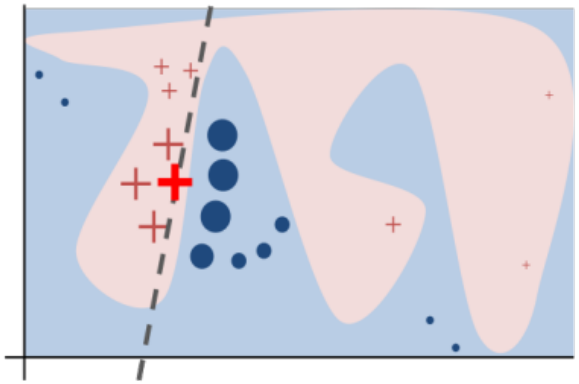
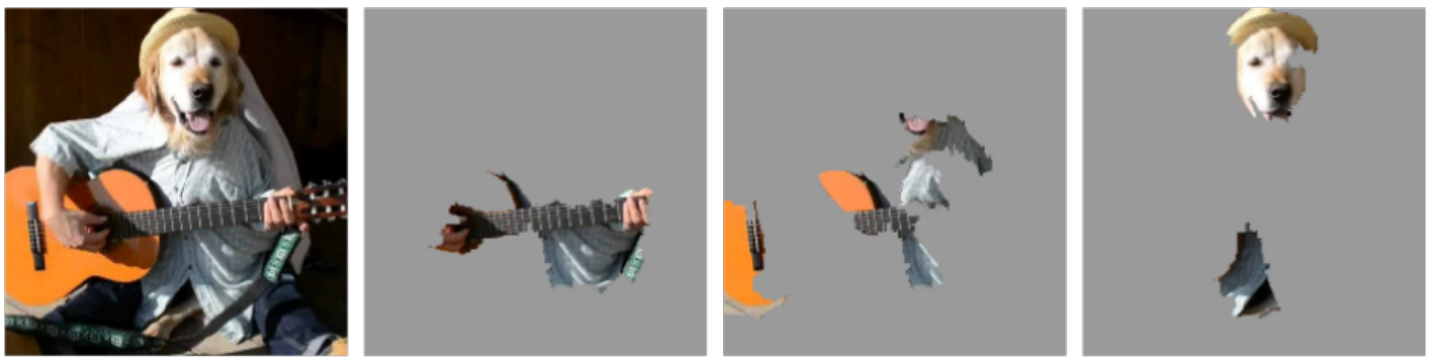


Figure 3: Toy example to present intuition for LIME. The black-box model's complex decision function f (unknown to LIME) is represented by the blue/pink background, which cannot be approximated well by a linear model. The bright bold red cross is the instance being explained. LIME samples instances, gets predictions using f, and weighs them by the proximity to the instance being explained (represented here by size). The dashed line is the learned explanation that is locally (but not globally) faithful.

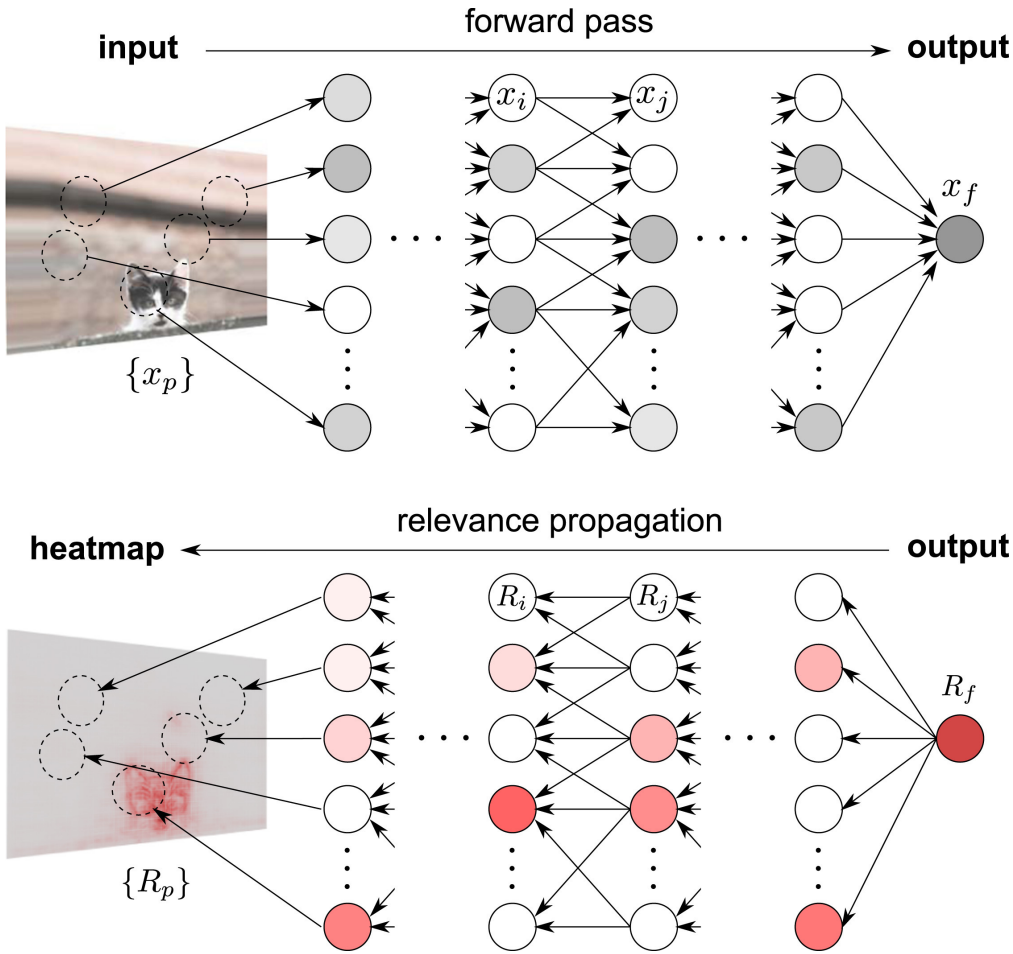


(a) Original Image (b) Explaining *Electric guitar* (c) Explaining *Acoustic guitar* (d) Explaining *Labrador*

Figure 4: Explaining an image classification prediction made by Google's Inception network, highlighting positive pixels. The top 3 classes predicted are "Electric Guitar" ($p = 0.32$), "Acoustic guitar" ($p = 0.24$) and "Labrador" ($p = 0.21$)

Ribeiro et al, 'Why Should I Trust You?' Explaining the Predictions of Any Classifier,
`https://arxiv.org/pdf/1602.04938v1.pdf`

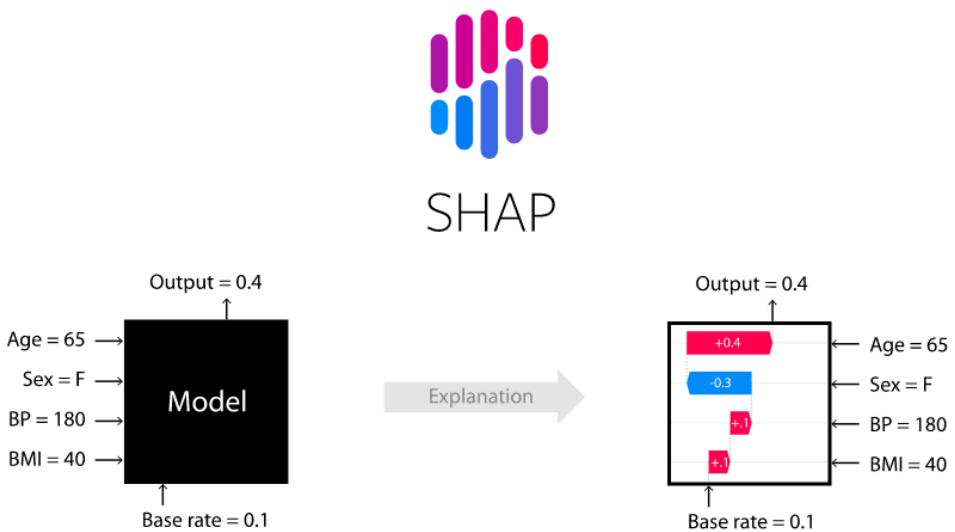
Layer-wise relevance propagation



Bach et al, On Pixel-Wise Explanations for Non-Linear Classifier Decisions by Layer-Wise Relevance Propagation `https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0130140`, Fig from Montavon et al, 2017

SHAP

- Shapley values are a game-theoretical model for assessing feature importance
- Various instances (and approximations) for different ML models – for images there are links to integrated gradients
- Post-hoc, algorithm-agnostic method for assigning relevance to input features



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Adversarial attacks

Saliency beyond images: Graphs

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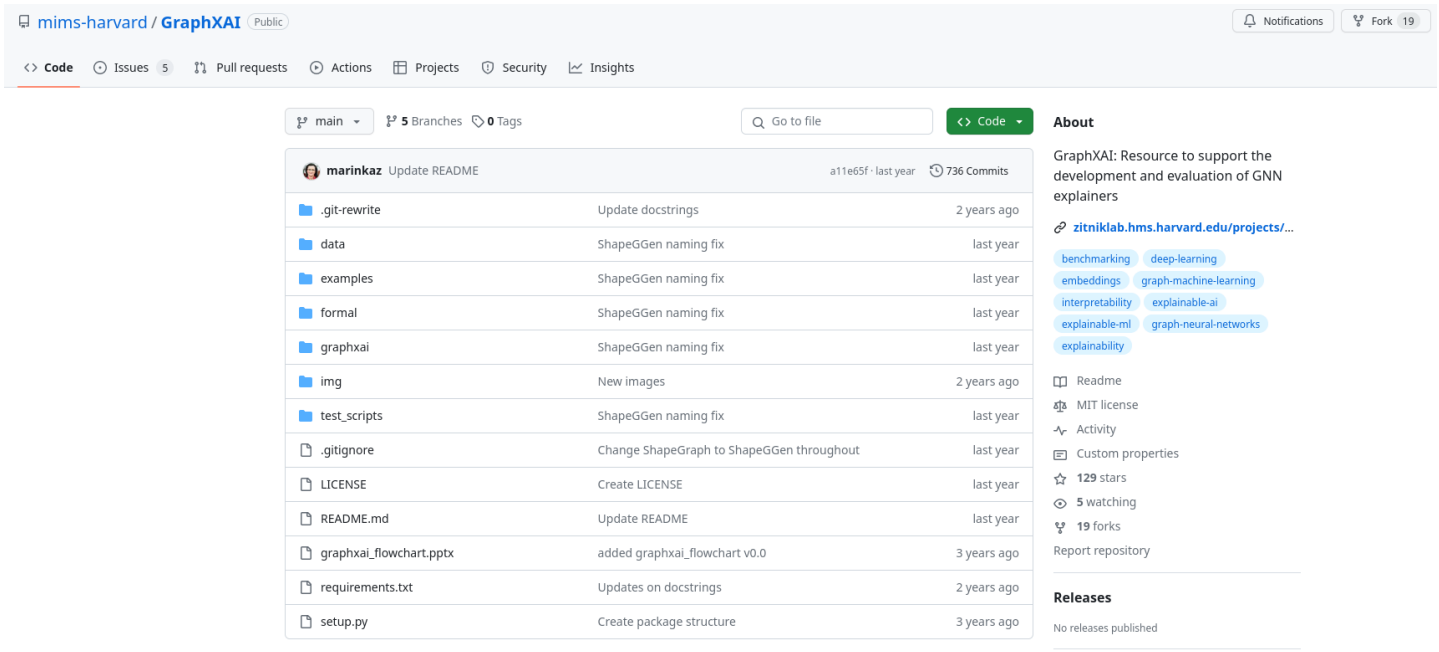
Abstract

As explanations are increasingly used to understand the behavior of graph neural networks (GNNs), evaluating the quality and reliability of GNN explanations is crucial. However, assessing the quality of GNN explanations is challenging as existing graph datasets have no or unreliable ground-truth explanations. Here, we introduce a synthetic graph data generator, SHAPEGGEN, which can generate a variety of benchmark datasets (e.g., varying graph sizes, degree distributions, homophilic vs. heterophilic graphs) accompanied by ground-truth explanations. The flexibility to generate diverse synthetic datasets and corresponding ground-truth explanations allows SHAPEGGEN to mimic the data in various real-world areas. We include

Saliency beyond images: Graphs

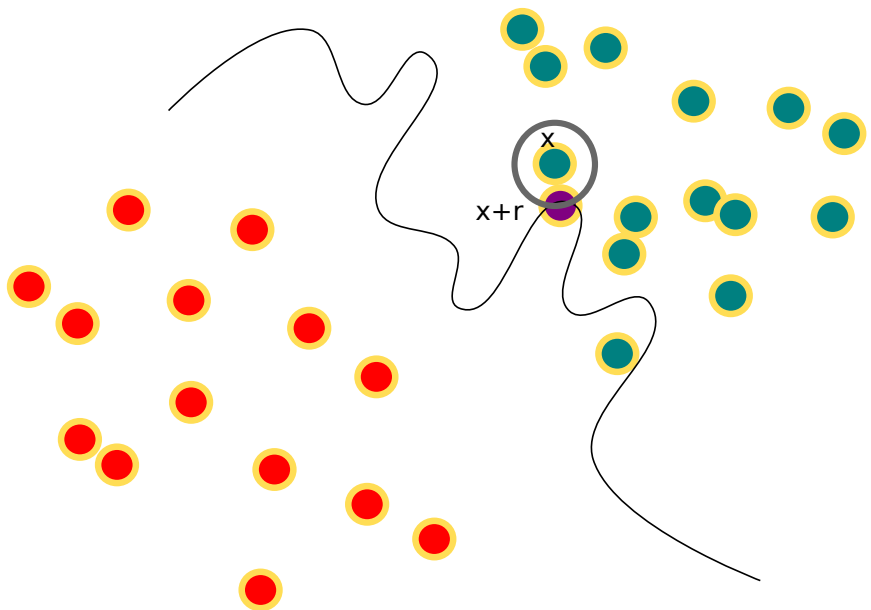


Adversarial attacks

- Given: Image $x \in \mathbb{R}^{H \times W}$, classifier $f : \mathbb{R}^{H \times W} \to \{0, \ldots, L\}$, and a target class label l
- An adversarial attack on the image x towards the class l: Find the image noise $r \in \mathbb{R}^{H \times W}$:

$$\text{minimize} \|r\|^2 \text{ while } \begin{array}{l} f(x+r) = l \\ x + r \in [0,1]^{H \times W} \end{array}$$

- **Question:** How would you find such an example?



Szegedy et al, Intriguing properties of neural networks , https://arxiv.org/abs/1312.6199, 2014

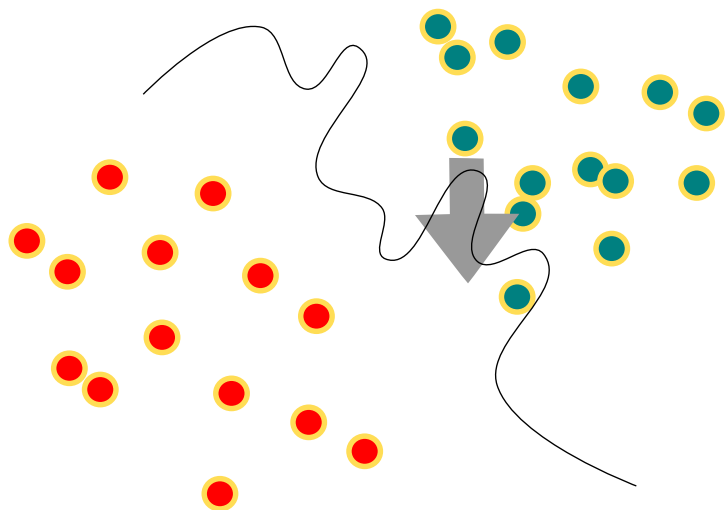
Adversarial attacks: Fast gradient sign method (FSGM), 2015

- The fast gradient sign method obtains adversarial examples via "noise"

$$r = \varepsilon \text{sign}(\nabla_x \mathcal{L}(\theta, x, y))$$

for loss $\mathcal{L}$, image x, correct label y, model parameters θ.

- In other words – move away from the correct class by maximizing its loss (linear approximation)



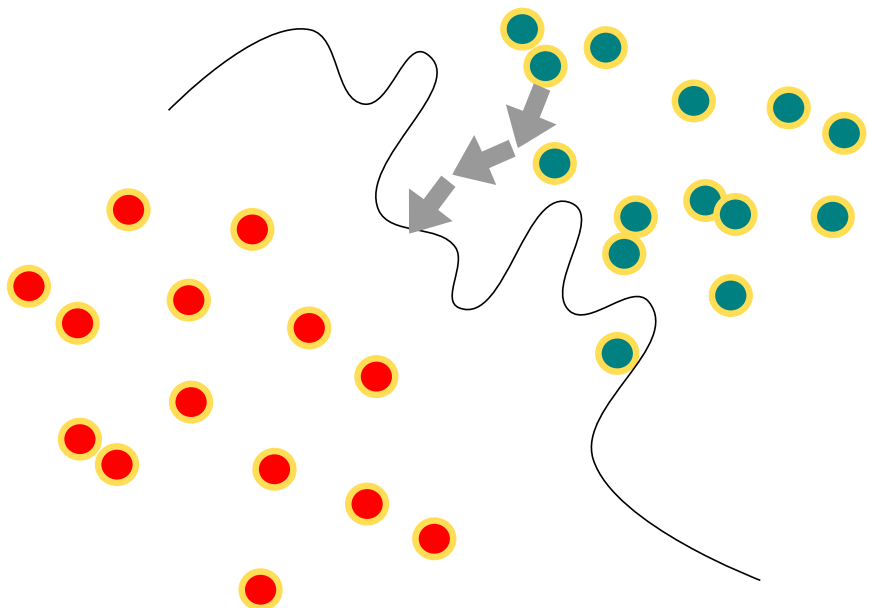
Goodfellow et al, Explaining and Harnessing Adversarial Examples, `https://arxiv.org/abs/1412.6572`

Adversarial attacks: Basic Iterative Method (BIM)

- Let go of the linear approximation by iterative attacking

$$x_0^{adv} = x, x_{n+1}^{adv} = \mathrm{Clip}(x_n^{adv} + \alpha \mathrm{sign}(\nabla_x \mathcal{L}(\theta, x, y))),$$

where Clip cuts off image values that go outside $[0, 255]$, and α is a step (e.g. 1).



Kurakin et al, Adversarial Examples in the Phyiscal World, `https://arxiv.org/pdf/1607.02533.pdf`, 2017

The XAI debate: XAI criticism

Saliency maps:
The # 1 hate object of explainable AI

https://www.thelancet.com/journals/landig/article/PIIS2589-7500(21)00208-9/fulltext#bib16

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The false hope of current approaches to explainable artificial intelligence in health care

Marzyeh Ghassemi, PhD • Luke Oakden-Rayner • Andrew L Beam, PhD

Open Access • Published: November, 2021 • DOI: https://doi.org/10.1016/S2589-7500(21)00208-9 •

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Summary
Introduction
Current approaches to explainable AI
What are explanations for?
Better and more

Summary

The black-box nature of current artificial intelligence (AI) has caused some to question whether AI must be explainable to be used in high-stakes scenarios such as medicine. It has been argued that explainable AI will engender trust with the health-care workforce, provide transparency into the AI decision making process, and potentially mitigate various kinds of bias. In this Viewpoint, we argue that this argument represents a false hope for explainable AI and that current explainability methods are unlikely to achieve these goals for patient-level decision support. We provide an overview of current explainability techniques and highlight how various failure cases can cause problems for decision making for individual patients. In the absence of suitable explainability methods, we advocate for rigorous internal and external validation of AI models as a more direct means of achieving the goals often associated

Saliency maps:
The # 1 hate object of explainable AI

Deep learning saliency maps do not accurately highlight diagnostically relevant regions for medical image interpretation

Adriel Saporta, Xiaotong Gui, Ashwin Agrawal, Anuj Pareek, Steven QH Truong, Chanh DT Nguyen, Van-Doan Ngo, Jayne Seekins, Francis G. Blankenberg, Andrew Y. Ng, Matthew P. Lungren, Pranav Rajpurkar

doi: https://doi.org/10.1101/2021.02.28.21252634

This article is a preprint and has not been peer-reviewed [what does this mean?]. It reports new medical research that has yet to be evaluated and so should *not* be used to guide clinical practice.

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Abstract | **Full Text** | Info/History | Metrics | Preview PDF

Abstract

Deep learning has enabled automated medical image interpretation at a level often surpassing that of practicing medical experts. However, many clinical practices have cited a lack of model interpretability as reason to delay the use of "black-box" deep neural networks in clinical workflows. Saliency maps, which "explain" a model's decision by producing heat maps that highlight the areas of the medical image that influence model prediction, are often presented to clinicians as an aid in diagnostic decision-making. In this work, we demonstrate that the most commonly used saliency map generating method, Grad-CAM, results in low performance for 10

Posted March 02, 2021.

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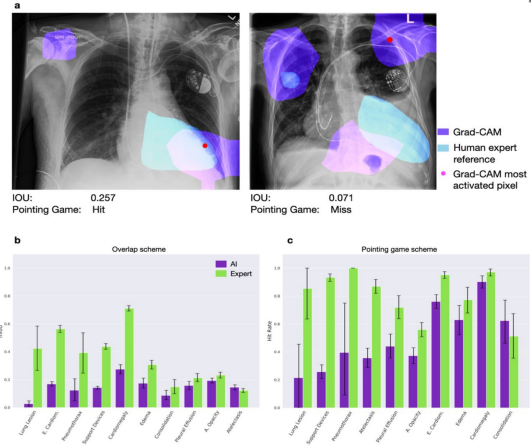

Question: What is happening?

Saliency maps:
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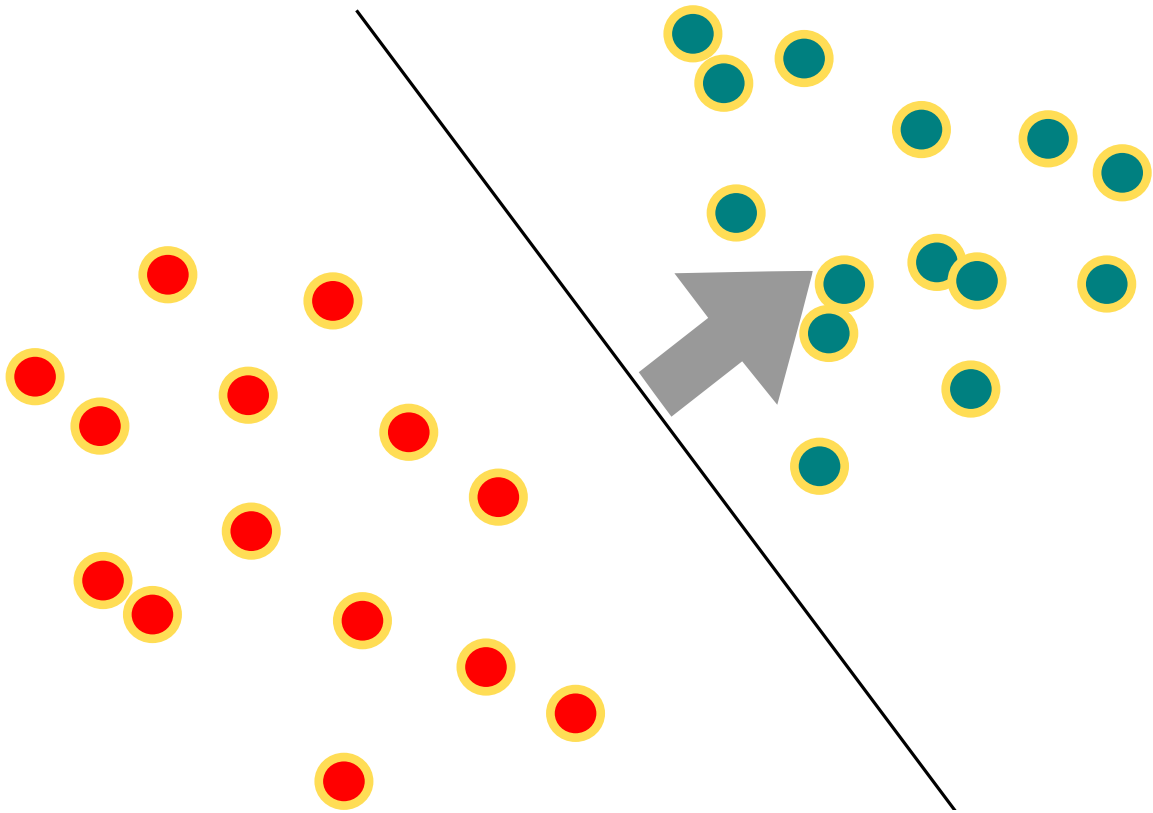

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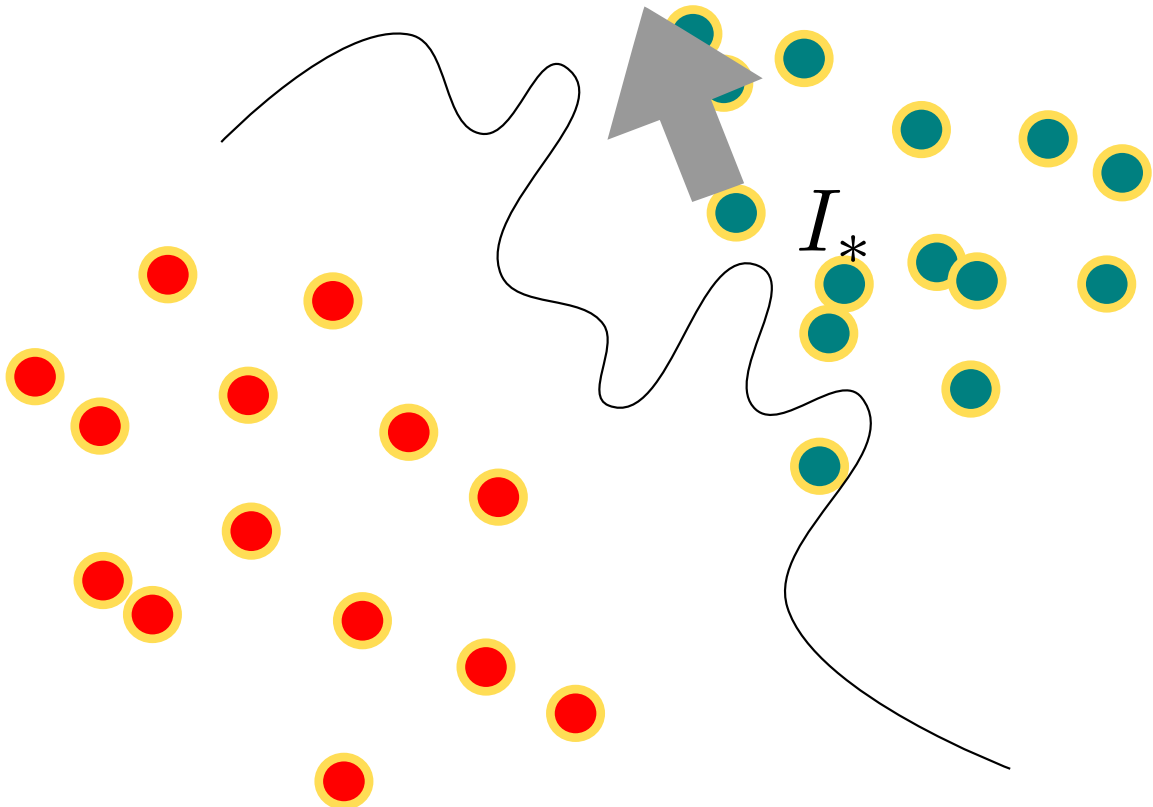

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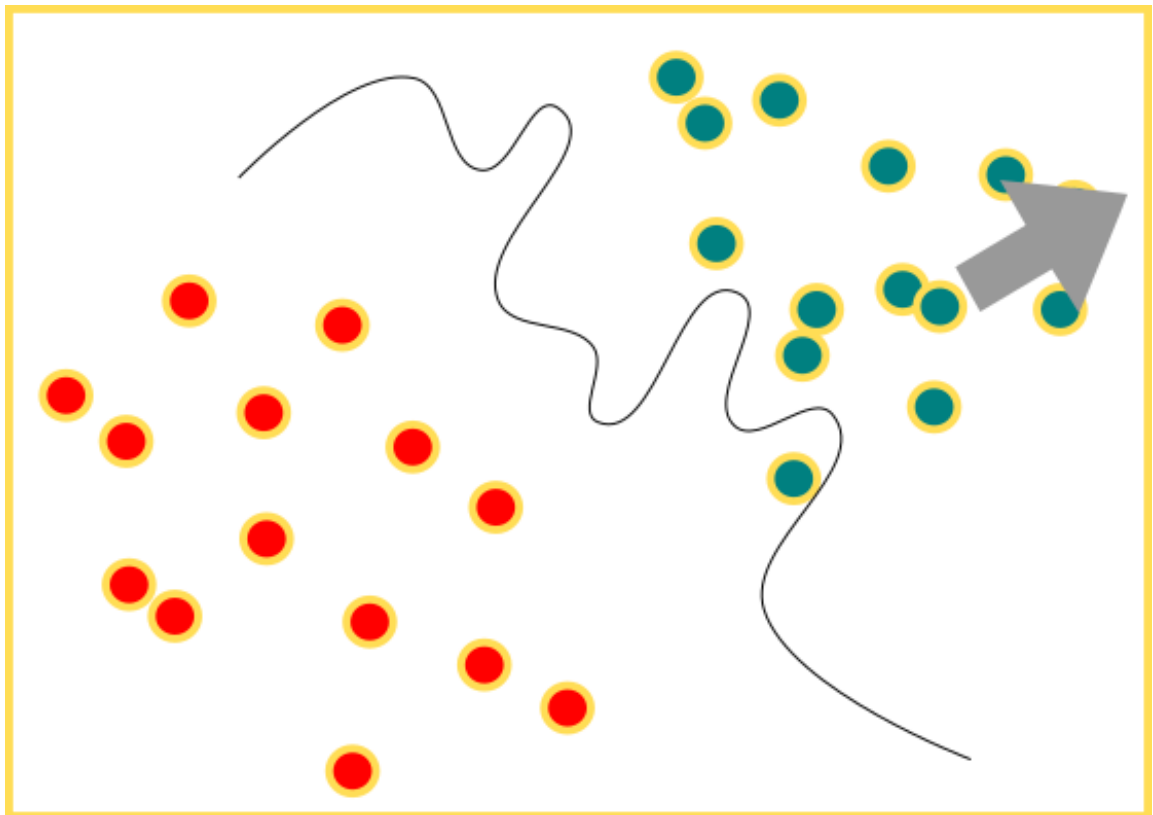

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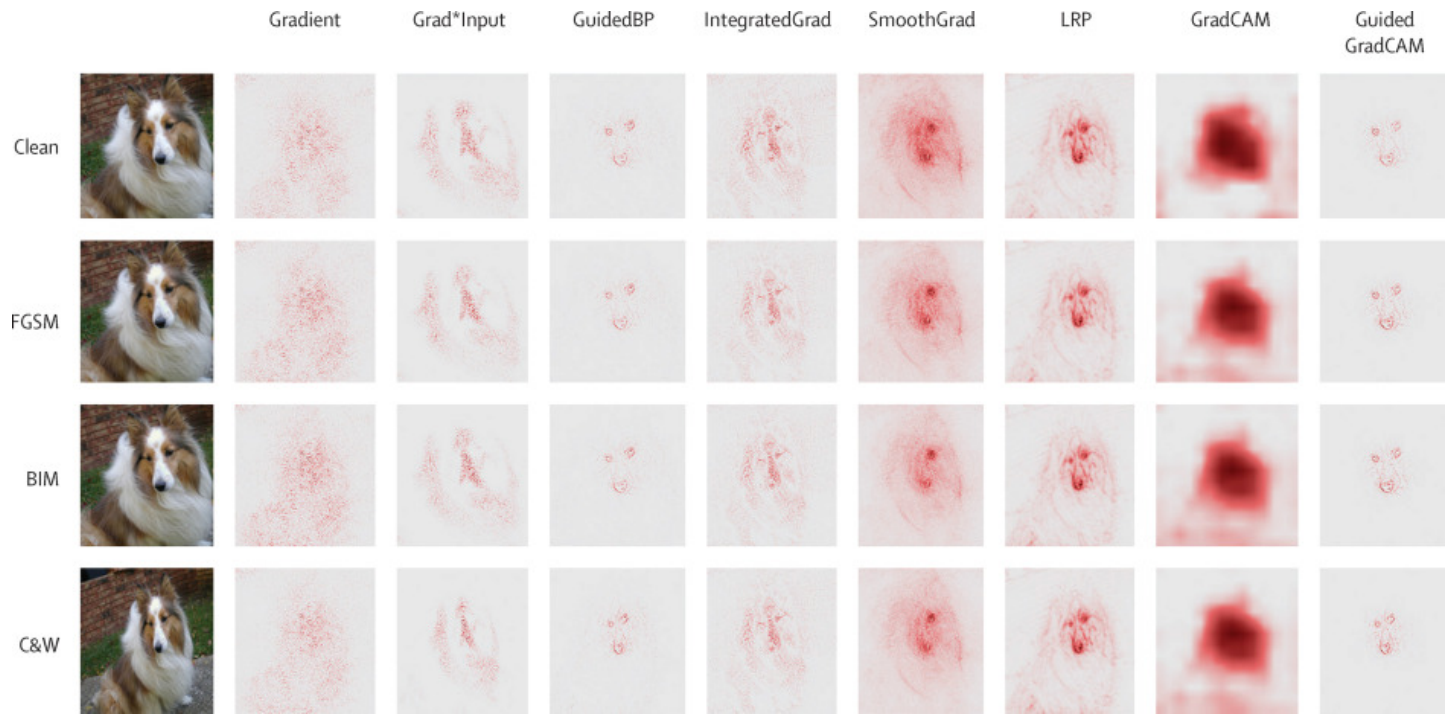

Although the prediction is changed using adversarial attacks, the saliency map looks unchanged!

Question: What is happening?

Gu and Tresp, arXiv, 2019

Saliency maps:
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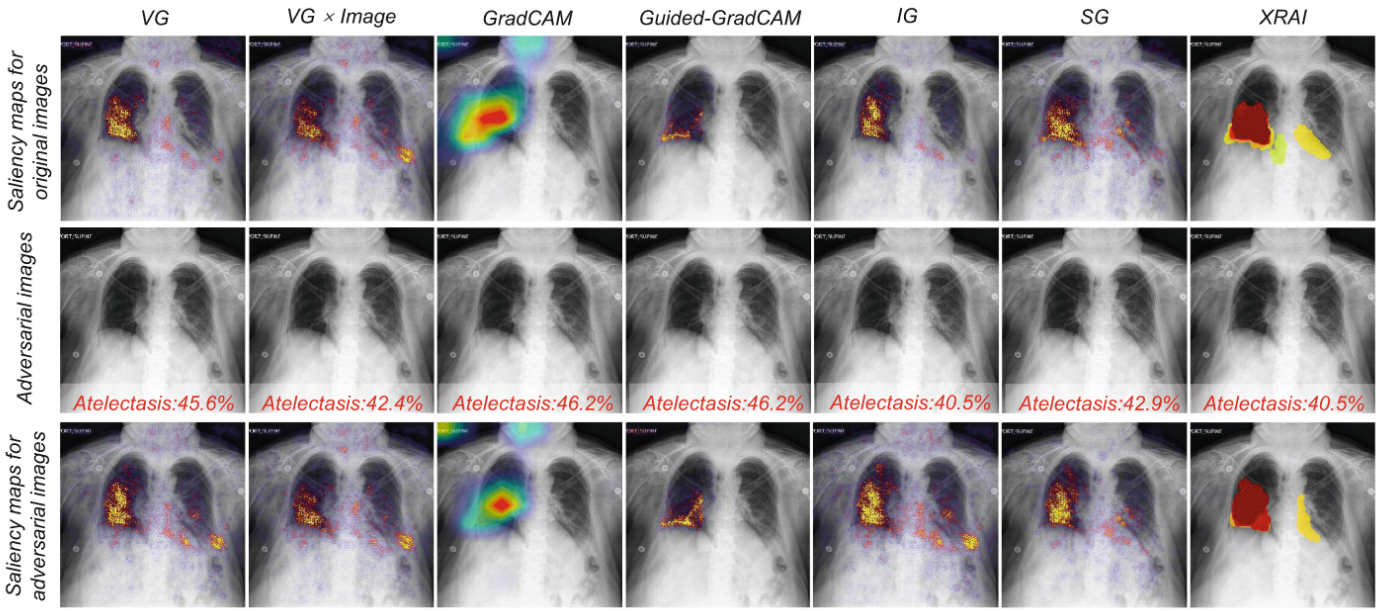

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Question: What is happening?

Zhang et al, Overlooked Trustworthiness of Saliency Maps, MICCAI'22

Saliency maps:
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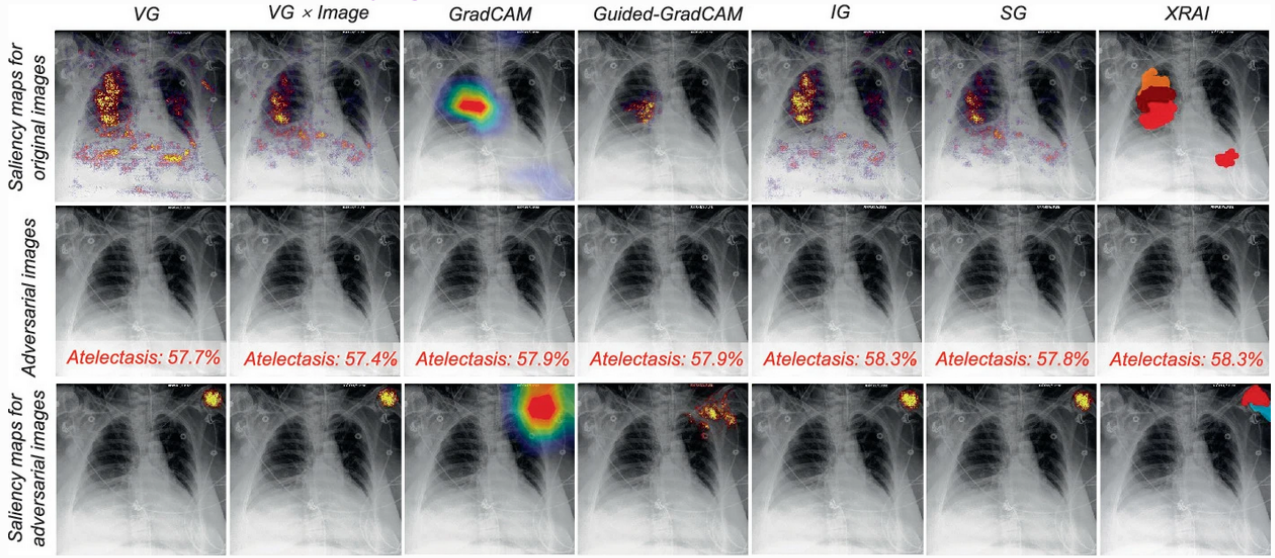
From: Overlooked Trustworthiness of Saliency Maps



An example of tampering with saliency maps without changing model predictions. The original prediction probabilities of Atelectasis is 57.7%.

An adversarial image with a customized saliency map can be "ordered"!

Question: What is happening?

Zhang et al, Overlooked Trustworthiness of Saliency Maps, MICCAI'22

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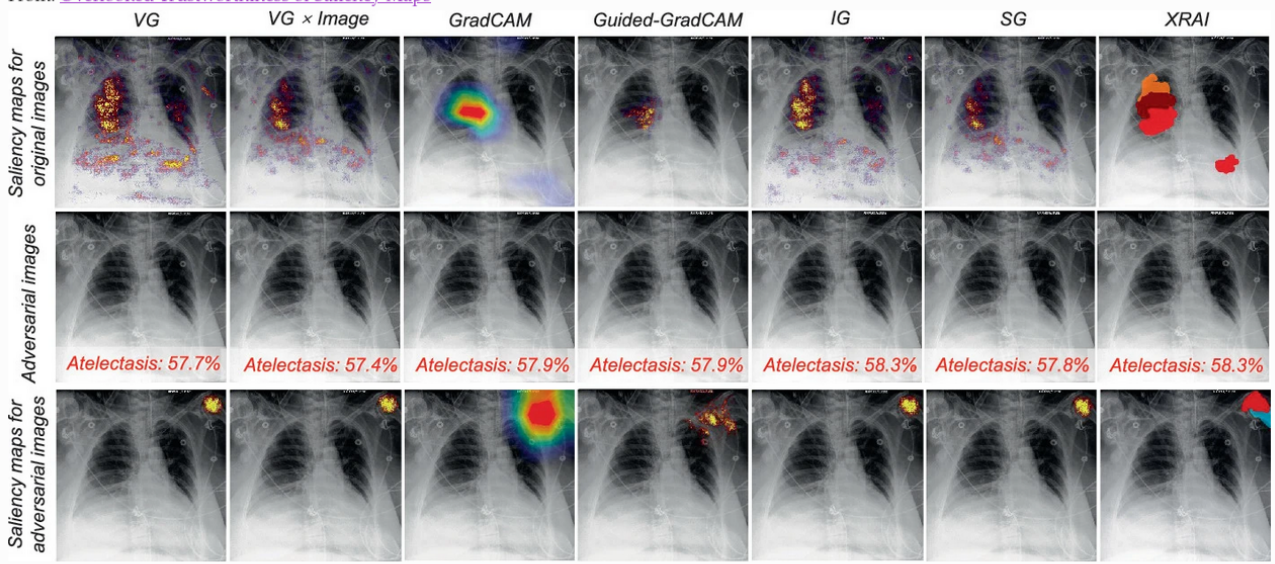
From: Overlooked Trustworthiness of Saliency Maps



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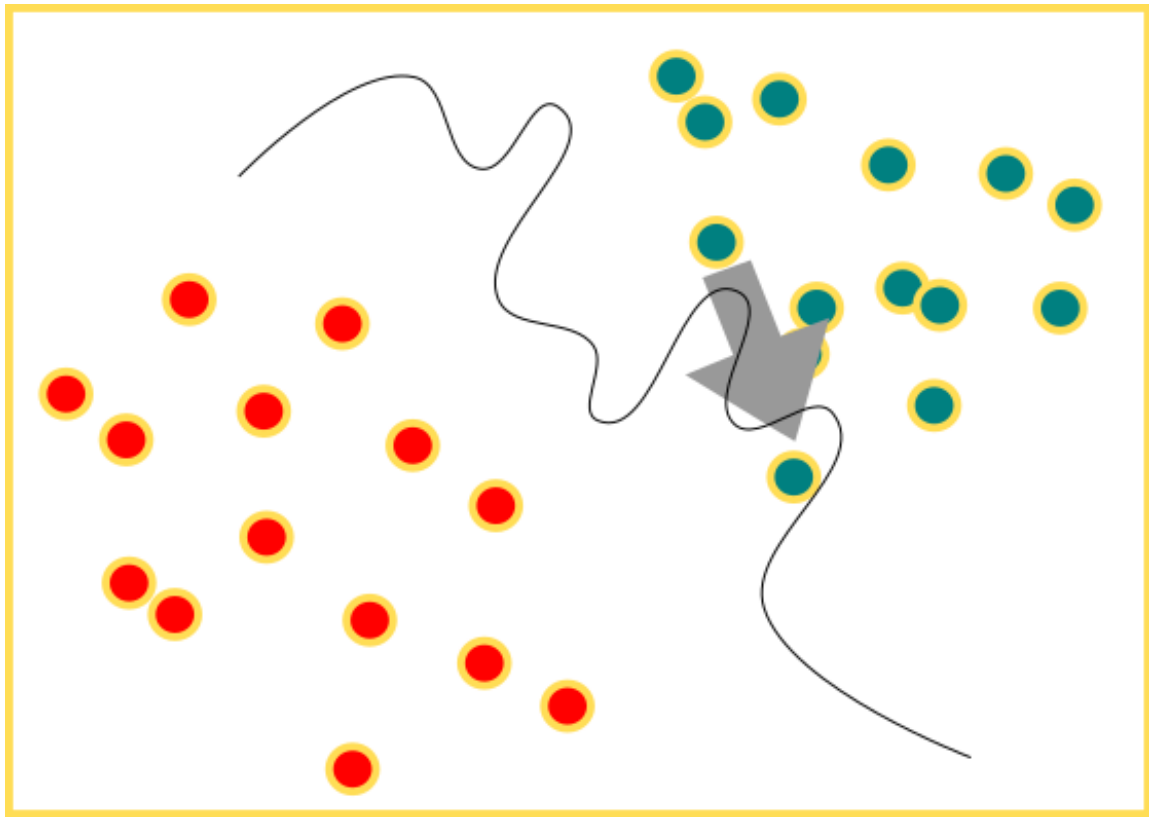
An adversarial image with a customized saliency map can be "ordered"!

Question: What is happening?

$$x_s = \underset{x'}{\operatorname{argmin}} \{||s_k(x') - s_{target}||^2 + \gamma_2 \cdot KL(f(x'), f(x))\},$$

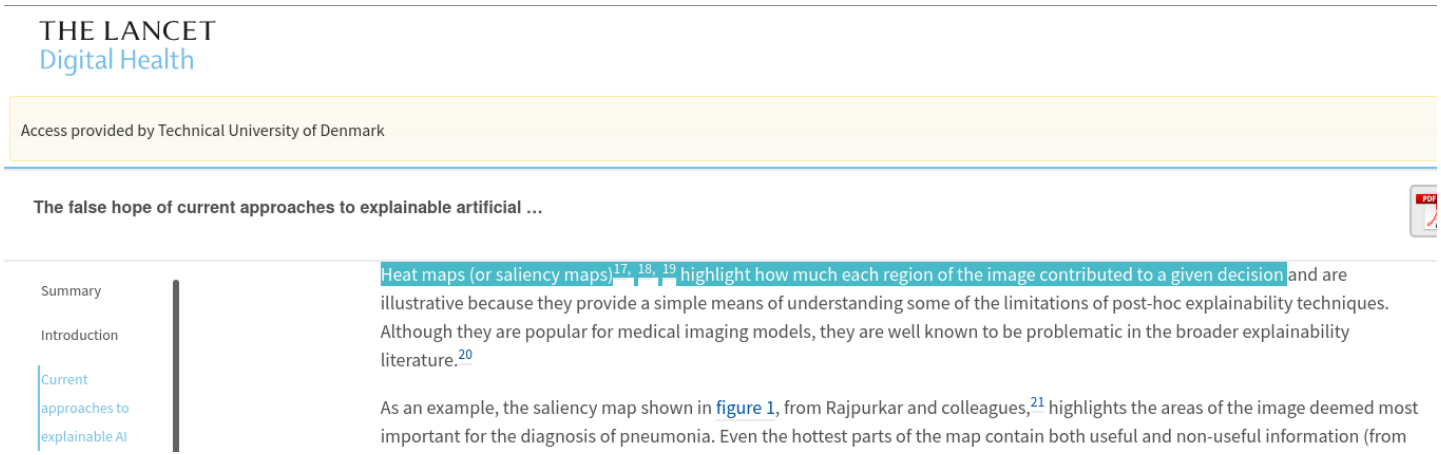
Zhang et al, Overlooked Trustworthiness of Saliency Maps, MICCAI'22

Saliency maps:
The # 1 hate object of explainable AI



Adversarial attack... we are just seeing the effect of a smooth gradient.

Saliency maps:
The # 1 hate object of explainable AI



THE LANCET
Digital Health

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The false hope of current approaches to explainable artificial ...

Summary

Introduction

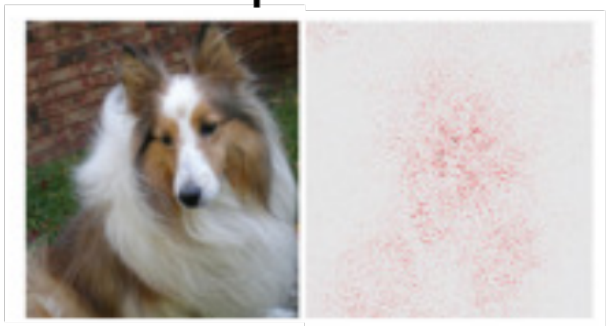
Current approaches to explainable AI

Heat maps (or saliency maps)[17, 18, 19] highlight how much each region of the image contributed to a given decision and are illustrative because they provide a simple means of understanding some of the limitations of post-hoc explainability techniques. Although they are popular for medical imaging models, they are well known to be problematic in the broader explainability literature.[20]

As an example, the saliency map shown in figure 1, from Rajpurkar and colleagues,[21] highlights the areas of the image deemed most important for the diagnosis of pneumonia. Even the hottest parts of the map contain both useful and non-useful information (from

Saliency maps:
The # 1 hate object of explainable AI

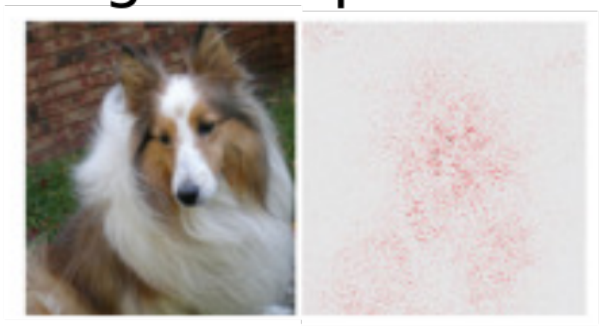
What contributed
to the prediction?



Saliency maps:
The # 1 hate object of explainable AI

~~What contributed to the prediction?~~

How do I efficiently change the prediction?



Group Discussion

Please discuss in groups:

- Suggest a more (mathematically) precise requirement for what constitutes an explanation
- Suggest a validation scheme that can check whether a particular XAI method lives up to those requirements
- Analyze the pros and cons of your particular requirement
- Put a brief summary on one of the slides from the announcement on Learn

Take-away message

- Saliency maps and related methods show you exactly what they are asked to show – but they do not all do the same
- ... so the user still needs to know what they do...
- ... in other words, explainable AI still cannot stand without an explanation.

Take-away message

- Saliency maps and related methods show you exactly what they are asked to show – but they do not all do the same
- ... so the user still needs to know what they do...
- ... in other words, explainable AI still cannot stand without an explanation.

Also:

- The "problems" associated with saliency maps are really caused by unwanted out-of-sample behavior of our models – not by the explanation

For next time

Please read the two papers linked in the course overview – we will both discuss these and go through them in detail.